

12-Week Practice Plan — Class 5 Mathematics Olympiad

Built around the SOF IMO pattern (50 Q / 60 marks / 60 min — see [02-exam-pattern.md](#)). Same three-phase shape as the Class 4 plan: build core skills topic by topic, add pure logical reasoning, then stress-test with full timed papers — with one extra core week, since Class 5 adds two genuinely new topic areas over Class 4.

Suggested pace: **3 short sessions a week, ~30–40 minutes each.**

Phase 1 — Core Skills (Weeks 1–8)

Week	Topic	Folder
1	Number Sense & Numerals (7–8 digit numbers, lakhs/crores, Roman numerals, rounding)	assignments/01-number-sense/  built
2	Four Operations (large numbers, multi-digit, word problems)	assignments/02-four-operations/
3	Factors, Multiples, HCF/LCM & Number Patterns	assignments/03-factors-multiples-patterns/
4	Fractions & Decimals	assignments/04-fractions-decimals/
5	Percentage, Ratio & Proportion, Profit/Loss & Simple Interest (new vs. Class 4)	assignments/05-percentage-ratio-profit-loss/
6	Measurement (Length, Weight, Capacity, Time, Temperature) & Money	assignments/06-measurement-money/
7	Geometry, Symmetry, Perimeter & Area	assignments/07-geometry-perimeter-area/
8	Data Handling	assignments/08-data-handling/

Weekly routine: attempt the assignment untimed first; mark it against the answer key; redo only the missed questions 2 days later.

Phase 2 — Logical Reasoning (Weeks 9–10)

Week	Topic	Folder
9	Patterns, Analogy & Classification, Alphabet Test, Coding–Decoding, Logical Sequence of Words, Puzzle Test	<code>assignments/09-reasoning-patterns-coding/</code>
10	Ranking Test, Direction Sense, Mirror & Water Images, Embedded Figures, Clock & Calendar	<code>assignments/10-reasoning-direction-figures/</code>

Note on figure-based items (mirror/water images, embedded figures): see the approach used in the Class 4 kit's `assignments/10-reasoning-direction-figures/images/` — real SVG diagrams, not text substitutes. Build these the same way here when Week 10 is filled in.

Phase 3 — Achievers / HOTS + Full Paper (Weeks 11–12)

Week	Activity
11	<code>assignments/11-achievers-hots/</code> — 2-and-3-step mixed problems pulling from every Phase-1 topic. Then attempt Sample Paper 1 (untimed, open-book) to find the gaps — see <code>sample-papers/README.md</code> .
12	Attempt further sample papers, each strictly timed to 60 minutes . After each paper, log every wrong answer in an error log (topic + why it was wrong) before moving to the next paper.

Only Sample Paper 1 is built out in this repo so far (full 50 questions, 60 marks, answer key) — build more in the same format as the plan progresses, the same way Class 4 grew from 1 sample paper to 6.

Progress tracker

- ☐ Week 1 — Number Sense (*assignment ready*)
- ☐ Week 2 — Four Operations
- ☐ Week 3 — Factors, Multiples, HCF/LCM & Patterns
- ☐ Week 4 — Fractions & Decimals
- ☐ Week 5 — Percentage, Ratio & Proportion, Profit/Loss & Simple Interest
- ☐ Week 6 — Measurement & Money
- ☐ Week 7 — Geometry, Perimeter & Area
- ☐ Week 8 — Data Handling
- ☐ Week 9 — Reasoning: Patterns/Analogy/Coding/Word Sequence/Puzzles
- ☐ Week 10 — Reasoning: Ranking/Direction/Mirror & Water Images
- ☐ Week 11 — Achievers/HOTS + Sample Paper 1 (*paper ready*)

- ☐ Week 12 — Further sample papers, timed

Optional extension

Once the core plan is comfortably done, the Class 4 kit's [Math Kangaroo-style stretch set](#) is equally suitable here — Math Kangaroo's grade bands overlap (the "Ecolier" level spans grades 3–4, "Benjamin" spans grades 5–6), so a Class 5 student is better matched to the *next* band up in puzzle difficulty rather than that exact set. Worth building a Class-5-pitched version of it as a future addition.